Representational Measurement of Similarity: The Additive Difference Model, Revisited

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In dieser Ausarbeitung geht es darum, wie man eine Metrik auf ordinalen Ähnlichkeitsdaten herleiten kann, die zusätzlich additiv ist und durch das additive difference model repräsentiert werden kann. Das Hauptresultat ist, dass die l^p Metrik die einzige additive-difference Metrik, die additiv auf Liniensegmenten ist, was eng mit der Homogenität dieser Metrik zusammenhängt. Zum Herleiten dieses Hauptresultates werden die Grundelemente von Messtheorie, sowie von proximity measurement mithilfe von geometrische Modellen eingeführt. Diese Ausarbeitung basiert auf dem Paper Tversky and Krantz, 1970 und den Büchern Suppes, 2014 und Breinbauer, 1976.

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1 Introduction

Die Grundlegende Idee von Proximity measurement ist es, jeweils zwei Stimulipaare auf deren Ähnlichkeit zu vergleichen. Zum Beispiel könnten in einem Experiment Vier Farben a, b, c, d zusammen mit der Frage "Sind sich die Farben a und b ähnlicher als die Farben c und d" Präsentiert werden. Die Aufgabe wäre dann, immer die Stimulipaare nach Ihren Ähnlichkeiten zu sortieren. Diese Ähnlichkeitsvergleiche will man anschließend durch eine Metrik repräsentieren. Dies macht man, indem man fordert, dass wenn zwei Stimuli a und b ähnlicher sind, als zwei weitere Stimuli c und d, dass dann der metrische Abstand zwischen a und b kleiner ist als der Abstand zwischen c und d. In der Sprache der Messbarkeitstheorie heißt das, man möchte ordinale Daten mithilfe einer ordinalen Skala repräsentieren, die metrische Eigenschaften erfüllt. Da im weiteren die Grundlegenden Skalen noch öfters vorkommen, definieren wir diese hier in der Einleitung kurz und intuitiv, ohne die genaue mathematische Definition zu nennen.

Definition 1.1 (Skalentypen)**.**

Die Skala ist eine Eigenschaft von Messdaten und enthält Information über deren Struktur. Dabei wird zwischen Vier Grundlegenden Skalen unterschieden, die sich jeweils mathematisch durch die zulässigen Operationen und die zugrundeliegende Struktur unterscheiden.

1. Nominalskala: Skala bei der man nur 1-zu-1 Zuordnungen macht. Die zulässigen Transformationen sind alle bijektiven Abbildungen. Beispiele sind z.B. Geschlechter (in Fragebögen z.B. 1 weibl. 2 männl. 3 divers) oder Postleizahlen.

2. Ordinalskala Zusätzlich zu 1-zu-1 Zuordnungen hat man eine Reihenfolge. Daher sind die zulässigen Transformationen die streng monotonen Abbildungen. Beispiele sind Noten (1 bis 6) oder Erdbebenstärke.

3. Intervallskala Zusätzlich zur Reihenfolge, spielen die Abstände zweier Werte eine Rolle. Damit sind die zulässigen Transformationen affin lineare Abbildungen $f(x) = ax + b, a > 0$. Beispiele sind die Celsius und Fahrenheit Skala.

4. Verhältnisskala Zusätzlich zu den Abständen in der Intervallskala, gibt es eine feste Null. Daher sind die zulässigen Transformationen alle linearen Abbildungen $f(x) = ax, a > 0$. Beispiele sind Länge oder die Kelvin Skala.

Da wir uns mit Ordinaldaten beschäftigen werden, spiele diese Skala und streng monotone steigende Abbildungen eine wichtige Rolle. Daher wird streng monoton steigend nur mit steigend bezeichnet.

In den Abschnitten 2 bis 4 geht es erstmal um die Mathematische Basis, die benötigt wird um die Hauptresultate aus Abschnitt 5 herzuleiten. Daher werden die wichtigsten Definitionen und Sätze in den Abschnitten 2 bis 4 erläutert aber nicht bewiesen. An ein paar Stellen sind die Feinheiten der Definitionen auch nicht so wichtig, dort wird dann auf den Anhang verwiesen, wo die genauen Definitionen von einigen Begriffen zu finden sind. Alle übersprungenen Beweise sind in Suppes, 2014 zu finden. Außer den Sätzen in Suppes, 2014 benötigen wir noch 2 Weitere Resultate die nicht Beweisen werden: Mulholland, 1949 und Aczél, 1966, diese beiden Paper sind auf Moodle zu finden.

2 Proximity Structures and Metrics

In diesem Kapitel geht es erstmal darum die Mathematische Basis Aufzubauen. Wie aus der Einleitung hervorgeht, wollen wir jeweils zwei Paare von Stimuli auf ihre Ähnlichkeit vergleichen. Seien a, b, c und d solche Stimuli. Dann bedeutet die Notation $(a, b) \succsim (c, d)$, dass Stimuli a und b sich unähnlicher sind, als c und d. Analog bedeutet $\sim$, dass die Ähnlichkeit zweier Paare gleich ist.
Die Idee von einer proximity structure, ist die Mathematische Definition dieser Idee: eine Relation $\succsim$ auf einer Menge A die die Grundeigenschaften einer Metrik erfüllt.

Definition 2.1 (factorial proximity structure)**.**
Suppose A is a set and $\succsim$ a quaternery relation on A, i.e., binary relation on $A \times A$. $\langle A, \succsim \rangle$ is a proximity structure iff the following axioms hold for all $a, b \in A$:

1. $\succsim$ is a weak ordering on $A \times A$, i.e., it is connected and transitive.
2. $(a, b) \succsim (a, a)$ whenever $a \neq b$.
3. $(a, a) \sim (b, b)$ (minimality).
4. $(a, b) \sim (b, a)$ (symmetry).

The structure is factorial iff

$$A = \prod_{i=1}^{n} A_i.$$

Remark 2.2 (Connectedness)*.* Connected means, that comparison of all distinct elements is possible, i.e., for a relation R on a set X it hold for all $x, y \in X$ with $x \neq y$ that either xRy or yRx.

Da wir die Relation $\succsim$ durch eine Metrik repräsentieren wollen, lohnt es sich einen Begriff dafür einzuführen. Die Idee ist das, was man sich intuitiv vorstellt: man will dass die Metrik die Struktur von $\langle A, \succsim \rangle$ erhält, und die einzige Struktur die wir gegeben haben, ist die Ordnung jeweils zweier Paare.

Definition 2.3 (Representability of $\succsim$ by a metric)**.**
A proximity structure $\langle A, \succsim \rangle$ is representable by a metric iff there exists a real-valued function δ on A such that:

1. $\langle A, \delta \rangle$ is a metric space, i.e., δ is a metric on A.
2. δ preserves the relation $\succsim$, i.e., for all $a, b, c, d \in A$

$$\delta(a, b) \geq \delta(c, d) \iff (a, b) \succsim (c, d).$$

3 Additive Difference Models and Structures

3.1 Additive Difference Model

Definition 3.1 (additive difference model)**.**

1. *Decomposability*: The distance between stimuli is a function of componentwise contributions.

$$\delta(a,b) = F[\psi_1(a_1,b_1),\dots,\psi_n(a_n,b_n)], \tag{1}$$

where F is an increasing [1] real-valued function in each of its n real arguments and ψ_i is a real-valued symmetric function satisfying $\psi_i(a_i,b_i) > \psi_i(a_i,a_i)$ when $a_i \neq b_i$ for all $i = 1,...,n$.

2. *Intradimensional subtractivity*: Each componentwise contribution is the absolute value of an appropriate scale difference.

$$\delta(a,b) = F[|\varphi_1(a_1) - \varphi_1(b_1)|,\dots,|\varphi_n(a_n) - \varphi_n(b_n)|], \tag{2}$$

where F is as in eq. (1) and ψ_i is replaced with $|\varphi_i(a_i)-\varphi_i(b_i)|$ for some real-valued function φ_i defined on A_i for $i = 1,...,n$.

3. *Interdimensional additivity*: The distance is a function of the sum of componentwise contributions.

$$\delta(a,b) = G\left[\sum_{i=1}^{n}\psi_i(a_i,b_i)\right], \tag{3}$$

where ψ_i are as in eq. (1) and G is an increasing function in one argument.

4. *Additive difference model*: If both, additivity and subtractivity are assumed, we get the additive difference model.

$$\delta(a,b) = G\left\{\sum_{i=1}^{n}F_i[|\varphi_i(a_i) - \varphi_i(b_i)|]\right\}, \tag{4}$$

where G and F_i, for $i = 1,\dots,n$, are all increasing functions in one argument.

[1] We use the terms increasing and decreasing to denote real-valued functions that are strictly monotonically increasing and decreasing, respectively.

3.2 Additive Difference Structure

3.3 Examples

Example 3.2 (p-exp Metric Family).
Sei das F im additive difference model (4) gegeben durch

$$F(t) = q \cdot (e^{t \cdot r} - 1)^p \qquad q, r > 0, p \geq 1 \tag{5}$$

Dann ist das Modell eine Metrik.

Proof.

1. Positivität: $\delta(a,b) = 0$ also $F(|a_i - b_i|) = 0$ für alle i, das ist nur der Fall wenn der Exponent Null ist, also $a_i = b_i$ gilt für alle i.
   Anderes herum gilt, wenn $a \neq b$ ist, dass dann $a_i \neq b_i$ für mindestens ein i, woraus folgt dass $F_i(|a_i - b_i|) > 0$, also auch $\delta(a,b) > 0$.
2. Symmetrie: Symmetrie kommt daher, dass $|a_i - b_i| = |b_i - a_i|$ gilt.
3. Dreiecksungleichung: Die Dreiecksungleichung kommt aus dem Satz von Mullholland Mulholland, 1949, S. 297. Das Paper von Mulholland ist auf Moodle zu finden.

□

Theorem 3.3 (Mulholland).

Die Ungleichung

$$f^{-1}\left[\sum_i f(|x_i + y_i|)\right] \leq f^{-1}\left[\sum_i f(|x_i|)\right] + f^{-1}\left[\sum_i f(|y_i|)\right].$$

gilt für alle $x, y \in \mathbb{R}^n, n = 1, 2, \ldots$ wann immer die folgenden drei Bedingungen erfüllt sind:

1. *f ist stetig und streng monoton steigend, und $f(0) = 0$.*
2. *f ist konvex auf $[0, \infty)$.*
3. *$\log f(e^x)$ ist konvex für $x \in (-\infty, \infty)$.*

Beweis, dass 3.2 die Bedingungen von Mulholland erfüllt.
Zu Punkt 1: Stetigkeit und $f(0) = 0$ ist klar. Für Monotonie gilt, $e^{rt} - 1$ ist steigend, Potenzieren auch und multiplizieren mit $q > 0$ auch, damit ist die Komposition davon auch steigend. Zum Punkt 2: Selbes Argument wie bei Monotonie, nur mit Konvexität und Komposition von konvexen Funktionen ist wieder konvex.

Zum dritten Punkt, dass $\log F(e^t)$ konvex ist. Es gilt

$$\log q(e^{re^x} - 1)^p = \log q + p \cdot \log(e^{re^x} - 1)$$

Addition und Multiplikation mit $p \geq 1$ erhält Konvexität, also reicht es Konvexität von $\log(e^{re^x} - 1)$ zu zeigen.

Das zeigen wir über die zweite Ableitung, denn wenn die zweite Ableitung strikt positiv ist, ist die Funktion konvex.

Zweifaches Ableiten liefert

$$\frac{re^{re^x+x}\left(r(-e^x)+e^{re^x}-1\right)}{\left(e^{re^x}-1\right)^2} \overset{!}{>} 0$$

Wir sehen der Nenner ist immer positiv und ungleich 0, das heißt wir können den Nenner weg lassen. Genauso ist der Faktor re^{re^x+x} immer positiv und kann somit auch weggelassen werden. Es bleibt also zu zeigen, dass

$$-re^x + e^{re^x} - 1 > 0$$
$$\iff e^{re^x} - re^x > 1$$

gilt.

Dies kann man in 2 Schritten zeigen: 1. die Linke Seite ist streng monoton steigend und 2. der Grenzwert für $x \to -\infty$ ist 1. Somit gilt die Ungleichung für alle $x \in \mathbb{R}$.

1. Für die Ableitung gilt:

   $$re^x \cdot e^{re^x} - re^x = re^x \cdot (e^{re^x} - 1) > 0$$

   da $e^x > 0$ gilt.

2. $$\lim_{x\to-\infty} e^{re^x} - re^x = 1 - 0 = 1$$

   Damit ist die Konvexität gezeigt, und die Behauptung folgt aus dem Satz von Mulholland.

□

Remark 3.4 (l^p Metrik $\in$ p-exp Metrik Familie).
Wenn wir in (5) für $q = r^{-p}$ einsetzen, erhalten wir im Grenzwert für $r \to 0$ die l^p Metrik. Das ist von Bedeutung, da dies mithilfe des Hauptresultates **??** zeigt, dass die Metriken der p-exp Familie keine Metriken mit additiven Segmenten sein können, obwohl man die l^p Metrik durch Grenzwertbildung aus dieser Familie darstellen kann. Das bedeutet, dass im Grenzwertprozess aus der Eigenschaft "additiv auf den Koordinatenachsen", die Eigenschaft "additiv auf beliebigen Liniensegmenten" wird.

Proof. Setzt man $q = r^{-p}$ erhält man:

$$r^{-p}(e^{rt}-1)^p = \left(\frac{e^{rt}-1}{r}\right)^p$$

Für $r \to 0$ gehen sowohl Nenner, als auch Zähler, gegen 0 (wir können den Grenzwert in die Potenz reinziehen, da die Potenzfunktion stetig ist) und wir können den Satz von L'Hospital anwenden. Das liefert:

$$\lim_{r\to0}\left(\frac{e^{rt}-1}{r}\right)^p = \left(\lim_{r\to0}\frac{e^{rt}-1}{r}\right)^p = \left(\lim_{r\to0}\frac{t\cdot e^{rt}-1}{1}\right)^p = t^p$$

□

Nicht jede Funktion, die durch das additive difference model dargestellt werden kann, ist eine Metrik.

4 Segmental Additivity

4.1 Segmentally Additive Metrics

Da Metriken an sich wenig Struktur bieten, lohnt es sich noch eine Eigenschaft mehr mehr zu fordern.
Diese Eigenschaft ist segmental additivity und beschreibt, dass es zu zwei Punkten immer eine Kurve gibt, die diese verbindet und die Abstände additiv sind, das heißt auf dem Segment die Dreiecksungleichung mit Gleichheit gilt. Für uns reicht es zu wissen, dass wenn b auf dem Segment zwischen a und c liegt, dass dann $\delta(a, b) + \delta(b, c) = \delta(a, c)$ gilt. Für die genaue Definition siehe Anhang Am Ende dieses Abschnittes sind Beispiele für Metriken mit additiven Segmenten angegeben.

Der Folgende Satz liefert eine (bis auf positive Skalierung) eindeutige Metrik die die Relation repräsentiert und zusätzlich zur Ordnung, die Eigenschaft segmental solvability erhält.

Theorem 4.1 (Representation theorem for segmentally additive proximity structures)**.** *Suppose $\langle A, \succsim \rangle$ is a segmentally additive proximity structure. Then $\langle A, \succsim \rangle$ is representable by a metric, and additionally, for all $a, b, c \in A$:*

1. *$\langle abc \rangle \iff \delta(a, b) + \delta(b, c) = \delta(a, c)$.*
2. *If δ' is another metric on A that satisfies the above conditions, then there exists $\alpha > 0$ such that $\delta' = \alpha\delta$ (δ is a ratio scale).*

Definition 4.2 (temp)**.**
Todo: Metric, open, convex, stetig ... definieren?

The definitions and statements from this section are taken out from (Busemann, 1955).

Definition 4.3 (curves and segments)**.**

1. Let δ be a metric on X. A parametric curve in $\langle X, \delta \rangle$ is a function γ that maps a closed interval $[a, b] \subset \mathbb{R}$ into X and is continuous. The parametric curve γ is said to join $\gamma(a)$ and $\gamma(b)$.
2. Let γ be a parametric curve in $\langle X, \delta \rangle$ with domain $[a, b]$. Its length $\mathrm{L}(\gamma)$ is the supremum of the sums
$$\sum_{i=1}^{n} \delta\left[\gamma(a_{i-1}), \gamma(a_i)\right]$$
over all finite sequences $a_0, a_1, \ldots, a_n$ such that
$$a = a_0 \leq a_1 \leq \cdots \leq a_n = b.$$
If $\mathrm{L}(\gamma) < \infty$, then γ is rectifiable.
3. A rectifiable curve is a segment iff its length is equal to the distance between its endpoints, i.e. $\mathrm{L}(\gamma) = \delta\,(a,\ b)$.
4. For $a \leq x \leq b$ we denote the length of the sub-curve $\gamma(t),\ a \leq t \leq x$ from a to x by $\mathrm{L}_a^x(\gamma)$. This is also called the *arc length* of the curve.

Remark 4.4 (continuity & monotonicity of sub length). For a rectifiable curve $\gamma(t),\ a \leq t \leq b$, the arc length $\mathrm{L}_a^x(\gamma)$ is a continuous, non-decreasing function of x.

Definition 4.5 (Metric space with additive segments)**.**
Let δ be a metric on X. $\langle X, \delta \rangle$ is a metric space with additive segments iff for any $x, y \in X$ there is a segment joining them.

4.2 Segmentally Additive Proximity Structures

Da wir diese Additivität aus der proximity structure herleiten wollen, müssen wir Eigenschaften definieren, aus der man die Additivität herleiten kann. Dazu benötigen wir Folgenden Begriff:

Definition 4.6 (ternary relation on proximity structures)**.**
For $a, b, c \in A$, the ternary relation $(a, b, c\rangle$ is said to hold iff for all $a', b', c' \in A$:

1. If $(a, b) \succsim (a', b')$ and $(a', c') \succsim (a, c)$, then $(b', c') \succsim (b, c)$
2. If either of the preceding inequalities is strict, then the resulting inequality is also strict.

Further the relation $\langle abc\rangle$ is said to hold if both $(a, b, c\rangle$ and $(c, b, a\rangle$ hold.

Die Bedingungen, die notwendig sind, um eine Darstellung durch eine Metrik mit additiven Segmenten zu ermöglichen, sind in der folgenden Definition und dem Satz danach zusammengefasst. Der zweite Teil der Definition wird erst später, für das Hauptresultat (Satz **??**) relevant, im Prinzip geht es darum dass man sich Grenzwerte und Cauchy Folgen anschauen kann, also der normale Begriff von Vollständigkeit.

Definition 4.7 (segmentally additive proximity structure)**.**
A proximity structure $\langle A, \succsim\rangle$, with a ternary relation $\langle\rangle$ defined in terms of $\succsim$, is segmentally additive if and only if the following two axioms hold for all $a, c, e, f \in A$:

1. If $(a, c) \succsim (e, f)$, then there exists $b \in A$ such that $(a, b) \sim (e, f)$ and $\langle abc\rangle$. (segmental solvability)
2. If $e \neq f$, then there exist $b_0, \ldots, b_n \in A$, such that $b_0 = a$, $b_n = c$, and $(e, f) \succsim (b_{i-1}, b_i)$ for $i = 1, \ldots, n$. (connectedness)

A segmentally additive proximity structure is complete if and only if the following two axioms hold for all $a, c, e, f \in A$:

3. If $e \neq f$, then there exist $b, d \in A$, with $b \neq d$, such that $(e, f) \succ (b, d)$. (Non-discreteness of distances)
4. Let $a_1, \ldots, a_n, \ldots$ be a sequence of elements of A, such that for all $e, f \in A$, if $e \neq f$, then $(e, f) \succsim (a_i, a_j)$ for all but finitely many pairs (i, j). Then there exists $b \in A$, such that for all $e, f \in A$, if $e \neq f$, then $(e, f) \succsim (a_i, b)$ for all but finitely many i. (Limits for Cauchy sequences)

4.3 Examples

Example 4.8 (Segmental Additivity of the Minkowski Metric)**.**
For $p \geq 1$ the Minkowski Metric is a metric with additive segments. The segments are all straight lines (i.e. the line segments between two arbitrary points).

Proof. Let $x, z \in \mathbb{R}^n$ and $0 \leq t \leq 1$. Then, for $y = (1-t)x + tz$ (y is on the line segment between z and x, iff y can be represented in this way) it holds that:

$$\begin{aligned}
d(x,y) + d(y,z) &= \left[\sum_{i=1}^{n} |x_i - (1-t)x_i - tz_i|^p\right]^{1/p} + \left[\sum_{i=1}^{n} |(1-t)x_i + tz_i - z_i|^p\right]^{1/p} \\
&= t\left[\sum_{i=1}^{n} |x_i - z_i|^p\right]^{1/p} + (1-t)\left[\sum_{i=1}^{n} |x_i - z_i|^p\right]^{1/p} \\
&= d(x,z).
\end{aligned}$$

□

5 Combining Segmental Additivity with the AD-Model

Theorem 5.1 (Necessary and sufficient conditions for additive difference metrics)**.** *Suppose $\langle A, \succsim \rangle$ is an additive-difference proximity structure such that f is the additive-difference representation from* **??***, i.e.,*

$$f(a,b) = \sum_{i=1}^{n} F_i(|\varphi_i(a) - \varphi_i(b)|).$$

Suppose further that the domain of each F_i is a real interval and $F_i(0) = 0$, $i = 1, \ldots, n$ (through transformation, since F_i are interval scales).

For $\langle A, \succsim \rangle$ to be compatible with a metric, and additionally be additive on the coordinate axes,

1. *it is necessary that there exists a continuous, convex function F, such that $F_i(x) = F(t_i x), t_i > 0$, for all x in the domain of F_i, $i = 1, \ldots, n$; and*
2. *sufficient that, in addition (to the conditions in 1.), the function $\log F(e^x)$ is convex.*

Remark 5.2 (Additivity on the Coordinate Axes)*.* Additivity on the coordinate axes means the following: If $a, b, c \in A$ differ only on one factor A_i, then the triangle inequality of the metric holds with equality. In mathematical terms:

$$a_j = b_j = c_j \text{ for all } j \neq i \text{ and } (a,c) \succsim (a,b), (b,c) \Rightarrow \delta(a,b) + \delta(b,c) = \delta(a,c)$$

In the following lemmas until the proof of this theorem is finished everything will be assumed as in theorem 5.1. Todo: checken bis wohin genau

Corollary 5.3 (Form von δ)**.**

Seien die Voraussetzungen wie in Satz **??***. Dann wird die Metrik δ dargestellt durch*

$$\delta(a,b) = F^{-1}\Big\{ \sum_{i=1}^{n} F(|\varphi_i(a_i) - \varphi_i(b_i)|) \Big\} \tag{6}$$

wobei F stetig, konvex und steigend ist.

Proof of theorem 5.1 1. According to the assumptions of the theorem, there exists a metric δ on A and an increasing function G, such that for all $a, b \in A$,

$$\delta(a,b) = G\left(\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right) \right).$$

Assume a, b, and c differ only in the i-th factor and $a|b|c$. Thus, we can use additivity on the i-th coordinate axis for a, b, c which yields

$$G\left(F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) + G\left(F_i\left(|\varphi_i(b_i) - \varphi_i(c_i)|\right)\right) = G\left(F_i\left(|\varphi_i(a_i) - \varphi_i(c_i)|\right)\right).$$

Defining $|\varphi_i(a_i) - \varphi_i(b_i)| =: x$ and $|\varphi_i(b_i) - \varphi_i(c_i)| =: y$ yields

$$G[F_i(x)] + G[F_i(y)] = G[F_i(x+y)].$$

Further, define $H_i(x) = G[F_i(x)]$. This reduces the above equation to $H_i(x) + H_i(y) = H_i(x+y)$ for all x, y in the domain of $F_i, 1 \leq i \leq n$. This means, that H_i must be linear and since the values are positive, the only monotonic solution to this equation is $H_i(x) = t_i x$, $t_i > 0$. Since G is increasing, it has an inverse G^{-1} satisfying

$$G^{-1}[H_i(x)] = G^{-1}[t_i x] = F_i(x)$$

where F_i is defined. Since G^{-1} is independent of i, the F_i can only differ in range and in the unit of the domain (i.e. the t_i which adjusts the domain where G^{-1} is defined). Without loss of generality, we can assume that the range of F_n is the largest. Since the range of F_n includes the range of F_i, we can set $F = F_n$ and obtain $F_i(x) = F_n(t_i x) = F(t_i x)$ for $t_i > 0$ and for every x in the domain of F_i. Continuity and convexity will be proven in the next lemmas. □

Todo: sagen dass conditions wie oben sind

Lemma 5.4.

$$F(\alpha+\beta+\gamma) + F(\gamma) \geq F(\alpha+\gamma) + F(\beta+\gamma).$$

Proof. In the triangle inequality,

$$F^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] + F^{-1}\left[\sum_{i=1}^{n} F(\beta_i)\right] \geq F^{-1}\left[\sum_{i=1}^{n} F(\alpha_i+\beta_i)\right],$$

let $\alpha_1 = \gamma, \alpha_2 = F^{-1}(F(\alpha+\gamma) - F(\gamma)), \alpha_j = 0$ for $3 \leq j \leq n$ and $\beta_1 = \beta, \beta_j = 0$ for $2 \leq j \leq n$. Hence,

$$\begin{aligned} F(\alpha+\beta+\gamma) &= F\left[F^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] + F^{-1}\left[\sum_{i=1}^{n} F(\beta_i)\right]\right] \\ &\geq \sum_{i=1}^{n} F(\alpha_i+\beta_i) \qquad \text{(by the triangle inequality)} \\ &= F(\beta+\gamma) + F(\alpha+\gamma) - F(\gamma). \end{aligned}$$

□

Lemma 5.5.
F is continuous.

Proof. Since F is increasing, all discontinuities are gaps. Real analysis yields, that real monotonic functions have at most countable many gaps. We will derive a contradiction from this.

Assume $F(\alpha)$ is the lower boundary of a gap, then there exists $\epsilon > 0$ such that for all $\delta > 0$, $F(\alpha+\delta) - F(\alpha) > \epsilon$. However, by lemma 5.4, for every $\beta > 0$,

$$F(\alpha+\beta+\delta) - F(\alpha+\beta) \geq F(\alpha+\delta) - F(\alpha) > \epsilon,$$

and so there is a gap at any $\alpha+\beta$, and F would have to be discontinuous everywhere. This is impossible, so F must be continuous. ☐

Lemma 5.6 (Characterization of Convexity)**.**
Suppose F is a strictly increasing function defined on the nonnegative reals. Then F is convex iff for all $\alpha, \beta, \gamma \geq 0$ the inequality from 5.4 holds, i.e.,

$$F(\alpha+\beta+\gamma) + F(\gamma) \geq F(\alpha+\gamma) + F(\beta+\gamma).$$

Proof.

Part 1 (⇒)**.**
Suppose, F is convex, and let $p = \alpha/(\alpha+\beta)$, then

$$\begin{aligned} p(\alpha+\beta+\gamma) + (1-p)\gamma &= \alpha+\gamma \\ (1-p)(\alpha+\beta+\gamma) + p\gamma &= \beta+\gamma. \end{aligned}$$

Hence, by convexity,

$$\begin{aligned} F(\alpha+\gamma) + F(\beta+\gamma) &= F\left[p(\alpha+\beta+\gamma) + (1-p)\gamma\right] + F\left[(1-p)(\alpha+\beta+\gamma) + p\gamma\right] \\ &\leq pF(\alpha+\beta+\gamma) + (1-p)F(\gamma) + (1-p)F(\alpha+\beta+\gamma) + pF(\gamma) \\ &= F(\alpha+\beta+\gamma) + F(\gamma). \end{aligned}$$

Part 2 (⇐)**.**
To prove the converse, we will use the continuity of F by lemma 5.5. Suppose, $\alpha > \beta$, then

$$\begin{aligned} F(\alpha) + F(\beta) &= F\left(\frac{\alpha-\beta}{2} + \frac{\alpha-\beta}{2} + \beta\right) + F(\beta) \\ &\geq 2F\left(\frac{\alpha-\beta}{2} + \beta\right) && \text{(by lemma 5.5)} \\ &= 2F\left(\frac{\alpha+\beta}{2}\right), \end{aligned}$$

which is equivalent to convexity for a continuous F.

☐

5.1 Preliminary lemma

Nur kurze zusammenfassung der Notation zur Übersicht Notation:

- γ: parametric curve, $\mathrm{L}(\gamma)$: length of curve, $\mathrm{L}_a^x(\gamma)$: length of subcurve from a to x
- δ_∞: ∞-metric, if no subscript then δ is the additive difference metric
- δ: metric, $\delta\,(a,\ b)$ metric distance between a and b, $\delta\,(a) := \delta\,(0,\ a)$
- $\mathrm{B}_\delta\,(r) := \{x \in \mathbb{R}^n : \delta\,(x) < r\}$: the open Ball with radius r around 0 wrt. the metric δ,
- $\mathrm{B}_\delta\,[r] := \{x \in \mathbb{R}^n : \delta\,(x) \leq r\}$: the closed Ball with radius r around 0 wrt. the metric δ,
- $\|x, y\|_\infty := \max_{i\leq n}\{|x_i - y_i|\}$: the maximum metric. We just use ∞ to shorten the notation, e.g. in $\mathrm{B}_\infty\,(r)$.
- $\mathrm{conv}(A)$: convex Hull of A.
- $\mathrm{extreme}(A)$: extreme Points of A.

In this section we will prove the main lemma, which is needed for the proof of the main result First, we will state the lemma, then give some remarks on the lemma and prove one statement needed for the lemma and then finally prove it. The proof is divided into 5 parts and a whole subsection which is devoted to showing a statement that is needed in the proof. Everything in this section will be assumed as in this Theorem. For example, if we write δ, this implies not any metric but the metric from the lemma with its properties. Moreover all topological terms(open, closed, continuous) are meant with respect to the natural topology of $\mathbb{R}^n$ (i.e. the topology induced by the δ_∞ and the euclidean metric), not the topology induced by δ or the induced topology on S. This will be especially important for the subsection on convexity of the balls. We will begin with the statement:

Lemma 5.7 (Main Lemma)**.**
Let S be an open convex subset of n-dimensional Euclidean space. Let δ be a metric on S that satisfies intradimensional subtractivity [2] with respect to a continuous function $F : \mathbb{R}^n_{\geq 0} \to \mathbb{R}_{\geq 0}$ and linear coordinates [3] in S. If $\langle S, \delta\rangle$ is a metric with additive segments, then δ is homogeneous, i.e., for any $x, z \in S$ and any real $0 \leq t \leq 1$,

$$\delta[x, x + t(z - x)] = t\,\delta(x, z).$$

Now we will continue with a remark that makes it possible to move the set such that $0 \in S$ by using the translation invariance of δ.

[2] i.e. δ takes the form $\delta\,(x,\ y) = F(|x_1 - y_1|, \ldots, |x_n - y_n|)$.

[3] Linear coordinates means that the vector representations are cartesian as opposed to, for example, polar, where each coordinate entry is not the value on the corresponding factor of the cartesian product.

Remark 5.8 (Translation invariance and domain of δ and F). (i) δ is translation-invariant, i.e. $\delta(x+y,\ z+y) = \delta(x,\ z)$:

$$\begin{aligned}\delta(x+y,\ z+y) &= F[|x_1+y_1-(z_1+y_1)|,\dots,x_n+|y_n-(z_n+y_n)|]\\ &= F[|x_1-z_1|,\dots,|x_n-z_n|]\\ &= \delta(x,\ z)\end{aligned}$$

(ii) Because δ is invariant to translation in its domain, we can move S and redefine the metric on the shifted set. We do this by defining $S' = \{s-x : s \in S\}$ for $x \in S$ and $\delta'(a,b) := \delta(a+x,\ b+x)$ for $a, b \in S'$. Since $0 \in S'$, we also can introduce the following shorthand Notation $\delta'(x') := \delta'(0,x')$ for $x' \in S'$. From now on, we will assume that $0 \in S$, because by the argument just provided, we can shift S and δ such that $0 \in S$. Thus we can proceed to proof the statement using the shorthand notation, by which we need to show that:

$$\delta(t \cdot z) = t \cdot \delta(z) \qquad \forall t \in [0,1] \text{ and } \forall z \in S'.$$

(iii) Since δ is defined by $\delta(x,\ y) = F(|x_1-y_1|,\dots,|x_n-y_n|)$ and δ is a metric, $F(x) = 0$ iff $x = 0$ must hold. Moreover the function F has to be defined on a set, where at least the "greatest possible difference" in coordinate i is a possible i-th argument. This means F is defined on a cartesian product of open, real intervals of the form $[0, a_i)$ where

$$a_i = \sup\{|x_i - y_i| : (x_1,\dots,x_n),(y_1,\dots,y_n) \in S\} \in \mathbb{R}_{>0} \cup \{\infty\}.$$

These a_i are these "greatest possible differences" on the i-th coordinate.

(iv) Moreover, due to the continuity of F and $|\cdot|$, the metric δ is continuous.

Before moving to the proof, we will fist show two statements that we will need right at the start. The first one asserts that the distance between a compact and a closed set is greater zero.

Todo: ref?

Lemma 5.9.
If $I \subset \mathbb{R}^n$ is compact, $Y \subset \mathbb{R}^n$ is closed and I and Y are disjoint, then there exists $\varepsilon > 0$ such that $\delta_\infty(p,\ y) > \varepsilon$ for all $p \in I$ and $y \in Y$.

Proof. We will prove this statement by contradiction. Assume that for all $\varepsilon > 0$ there exists $p \in I$ and $y \in Y$ such that $\delta_\infty(p,\ y) < \varepsilon$. This means, we can construct two sequences $p_n \in I$ and $y_n \in Y$ such that $\delta_\infty(p_n,\ y_n) \to 0$ as $n \to \infty$. I is compact, thus there exists a convergent subsequence p_{n_m} such that $p_{n_m} \to p \in I$ for $n \to \infty$. With the triangle inequality we get

$$\delta_\infty(p,\ y_{n_m}) \le \delta_\infty(p,\ p_{n_m}) + \delta_\infty(p_{n_m},\ y_{n_m}) \to 0 \text{ as } m \to \infty.$$

Hence, p is a limit point of Y, and since Y is closed, $p \in Y$ must hold which contradicts the assumption. □

The second one allows us to make arbitrary small δ-balls which we will combine with the statement before to assure a distance in δ and not δ_∞.

Lemma 5.10 (arbitrary small δ-Balls)**.**
Let $\varepsilon > 0$ such that $\mathrm{B}_\infty(\varepsilon) \subset S$. Then there exists $\alpha > 0$ such that $\mathrm{B}_\delta(\alpha) \subset \mathrm{B}_\infty(\varepsilon)$.

Proof. Since δ is a metric defined by $\delta(x, y) = F[|x_1 - y_1|, \ldots, |x_n - y_n|]$, we can define the j-th partial function of F by setting all other arguments to zero, i.e. $F^j(u_j) := F(0, \ldots, u_j, \ldots, 0)$.

F is strictly increasing in every argument implies that F^j is increasing and the continuity of F directly translates to F^j being continuous. Moreover, due to remark 5.8 (iii), $F^j(u_j) = 0$ if and only if $u_j = 0$ holds.

Together, this means that F^j is an increasing, continuous real function with $F^j(0) = 0$.

By setting $\alpha_j = F^j(\varepsilon)$ we get by monotonicity, that $F^j(u_j) < \alpha_j \implies u_j < \varepsilon$. Note, that $F^j(\varepsilon)$ is defined, because $\mathrm{B}_\infty(\varepsilon) \subset S'$ implies that ε is a possible distance for each coordinate. (See also remark 5.8 (iii). Using the a_i from the remark, we get that $\varepsilon < a_i$ holds for all i and thus $F^j(\varepsilon)$ is defined for all $j = 1, \ldots n$.)

Doing this for $j = 1, \ldots, n$ yields $\alpha_1, \ldots, \alpha_n$. By setting $\alpha = \min\{\alpha_1, \ldots, \alpha_n\}$ the above holds for every partial function F^j which yields the boundary for the ∞ Metric:

$$F^j(u_j) < \alpha \leq \alpha_j \implies u_j < \varepsilon \quad \forall j = 1, \ldots, n \implies \|u\|_\infty < \varepsilon.$$

Using the monotonicity of F we get

$$\delta(u) = F(u) < \alpha \implies F^j(u_j) < \alpha \implies \|u\|_\infty < \varepsilon.$$

□

Now, we will move to the proof of the main lemma. It is suggested to first read the proof to get the main ideas and a big picture understanding, and then read the next section which covers the convexity of spheres. First we will restate the lemma in the form, in which we will prove it.

Lemma (Restatement of the Lemma)**.**
Let S be an open convex subset of n-dimensional Euclidean space with $0 \in S$. Let δ be a metric on S that satisfies intradimensional subtractivity with respect to linear coordinates in S. If $\langle S, \delta \rangle$ is a metric with additive segments, then δ is homogeneous, i.e., for any $z \in S$ and any real $0 \leq t \leq 1$,

$$\delta(tz) = t\,\delta(z).$$

The proof consists of 4 parts. First we will find a radius such that all balls with this radius on line from 0 to z are in S. Then we will prove this lemma for $t = \frac{1}{m}$ for $m \in \mathbb{N}$ by proving the two inequalities. After this, we will expand this statement to the rationals and then use continuity to expand it to all real numbers between 0 and 1.

Proof.

Part 0 (Asserting a common distance for all points on the line segment).
Let $z \in S$. Since S is open, $\mathbb{R}^n \setminus S$ is closed. The set $I := \{tz : t \in [0,1]\}$ is compact, since continuous images($t \mapsto tz$) of compact sets ($t \in [0,1]$) are compactTodo: ref? . Thus, by lemma 5.9 there exists a distance $\varepsilon > 0$ such that $\delta_\infty(p,\ y) < \varepsilon$ for all $p \in I$ and $y \in \mathbb{R}^n \setminus S$. This means, for all points on the line from 0 to z the ball with radius ε is a subset of S, i.e. for all $t \in [0,1]$: $\mathrm{B}_\infty(tz,\ \varepsilon) \subset S$. By lemma 5.10, there exists $\alpha > 0$ such that $\mathrm{B}_\delta(\alpha) \subset \mathrm{B}_\infty(\alpha)$, and by translation invariance this holds for all points on the line: $\mathrm{B}_\delta(\alpha)\, tz \subset S$ for all $t \in [0,1]$. Now let $m \in \mathbb{N}$ such that $\frac{1}{m}\,\delta(z) < \alpha$. In particular, this means that $\mathrm{B}_\delta(\alpha)$ and therefore, $\mathrm{B}_\delta\left(\frac{1}{m}\,\delta(z)\right)$ is bounded (with respect to the ∞ metric).

Part 1 ($t\,\delta(z) \le \delta(tz)$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$).
Let $z^{(k)} := \frac{k}{m} \cdot z$ for $k = 0, \ldots, m$. $z^{(k)} \in S$ for all k due to convexity of S.

By applying translation invariance, we can calculate the distance between two consecutive points:

$$\delta\left(z^{(k-1)},\ z^{(k)}\right) = \delta\left(\frac{k-1}{m}z,\ \frac{k-1}{m}z + \frac{1}{m}z\right) = \delta\left(\frac{1}{m}z\right) \qquad \forall k = 1, \ldots, m.$$

Applying the triangle inequality $m-1$-times and using the above result yields

$$\delta(z) \le \sum_{k=0}^{m-1} \delta\left(z^{(k)},\ z^{(k+1)}\right) = m \cdot \delta\left(\frac{1}{m}z\right).$$

Finally, dividing the right side by m yields $\frac{1}{m}\,\delta(z) \le \delta\left(\frac{1}{m}z\right)$, which concludes the first part.

Part 2 ($t\,\delta(z) \ge \delta(tz)$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$).
Since δ is a metric with additive segments, there exists a segment from 0 to z with length $\delta(z)$. remark 4.4 yields, that the arc length is a continuous, non-decreasing function. This means, we can divide the segment into m subsegments with length $\frac{1}{m}\,\delta(z)$ and choose $y^{(0)}, \ldots, y^{(m)}$ equally spaced on the segment such that $\delta\left(y^{(i)},\ y^{(i-1)}\right) = \frac{1}{m}\,\delta(z)$ and $y^{(0)} = 0$ and $y^{(m)} = z$. Using this, we can define $x^{(i)} := y^{(i)} - y^{(i-1)}$. Translation invariance yields $\delta\left(x^{(i)}\right) = \delta\left(y^{(i)},\ y^{(i-1)}\right) = \frac{1}{m}\,\delta(z)$ and, due to the choice of m in part 0 $x^{(i)} \in S$ holds. Note, that this is still true, if we increase m.

Further, we can observe that $\frac{1}{m}z$ is a convex combination of $x^{(i)}$:

$$\sum_{i=1}^{m} \frac{1}{m}x^{(i)} = \sum_{i=1}^{m} \frac{1}{m}(y^{(1)} - y^{(0)} + y^{(2)} - y^{(1)} + \cdots + y^{(m)}) = \frac{1}{m}(-y^{(0)} + y^{(m)}) = \frac{1}{m}z$$

We showed before, that $\delta\left(x^{(i)}\right) = \frac{1}{m}\,\delta(z)$, which implies $x^{(i)} \in \mathrm{B}_\delta\left[\frac{1}{m}\,\delta(z)\right]$. By part 0 $\mathrm{B}_\delta\left(\frac{1}{m}\,\delta(z)\right)$ is bounded. Therefore we can apply lemma 5.11 from which we get that spheres are convex. This means, that the spheres are closed under convex combination, which implies that $\frac{1}{m}z \in \mathrm{B}_\delta\left[\frac{1}{m}\,\delta(z)\right]$ must hold. Thus $\delta\left(\frac{1}{m}z\right) \le \frac{1}{m}\,\delta(z)$ which concludes the second part of the proof.

Now, by combining part 1 and part 2, we have established the equality for $t = \frac{1}{m}$. Note, that the equality also holds for all $\mathbb{N} \ni m' > m$ since we only needed m to be big enough in part 2 for $x^{(i)}$ to be in S and for convexity of the spheres. Thus, by making m bigger, the crucial points in part 2 and therefore the equality are still true. Next, we will proceed to extending the equality from $t = \frac{1}{m}$ to $t = \frac{k}{m}$ for $k = 0, \ldots, m$.

Part 3 ($t\,\delta(z) = \delta(tz)$ for $t = \frac{k}{m}$ and $t \in \{0, \ldots, m\}$)**.**
For $z^{(1)}$ we already have $\delta\left(0,\ z^{(1)}\right) = \frac{1}{m}\,\delta(0,\ z)$. Now, for consecutive points, by translation invariance, it holds that

$$\delta\left(z^{(k-1)},\ z^{(k)}\right) = \delta\left(z^{(k-2)},\ z^{(k-1)}\right) = \cdots = \delta\left(z^{(1)},\ z^{(0)}\right) = \delta\left(z^{(1)},\ 0\right) = \frac{1}{m}\,\delta(0,\ z).$$

Using this and the triangle inequality, we get

$$\begin{aligned}
\delta(0,\ z) &\le \delta\left(0,\ z^{(1)}\right) + \delta\left(z^{(1)},\ z^{(2)}\right) + \cdots + \delta\left(z^{(m-1)},\ z^{(m)}\right) \\
&= n \cdot \frac{1}{n}\,\delta(0,\ z) \\
&= \delta(0,\ z) \\
\Longrightarrow \delta(0,\ z) &= \delta\left(0,\ z^{(1)}\right) + \delta\left(z^{(1)},\ z^{(2)}\right) + \cdots + \delta\left(z^{(m-1)},\ z^{(m)}\right)
\end{aligned}$$

For the distance $\delta\left(0,\ z^{(2)}\right)$, we get the following two inequalities:

1.
$$\delta\left(0,\ z^{(2)}\right) \le \delta\left(0,\ z^{(1)}\right) + \delta\left(z^{(1)},\ z^{(2)}\right) = \frac{2}{m}\,\delta(0,\ z)$$

2.
$$\begin{aligned}
\delta(0,\ z) &\le \delta\left(0,\ z^{(2)}\right) + \delta\left(z^{(2)},\ z\right) \\
\delta\left(0,\ z^{(2)}\right) &\ge \delta(0,\ z) - \underbrace{\delta\left(z^{(2)},\ z\right)}_{\le \sum_i \delta\left(z^{(2+i)},\ z^{(2+i+1)}\right)} \\
&\ge \delta(0,\ z) - \sum_{i=0}^{m-3} \delta\left(z^{(2+i)},\ z^{(2+i+1)}\right) \\
&= \delta(0,\ z) - \frac{m-2}{m}\,\delta\left(0,\ z^{(k)}\right) \\
&= \frac{2}{m}\,\delta(0,\ z)
\end{aligned}$$

These two inequalities yield $\delta\left(0,\ z^{(2)}\right) = \frac{2}{m}\,\delta(0,\ z)$. By repeating this for $\delta\left(0,\ z^{(k)}\right)$ for every $k = 2, \ldots, m-1$, we get the equality for every rational number for m big enough. However every rational number with a denominator smaller than m can be represented by another rational number with denominator greater than m. This yields, that the result is true for every rational t.

Part 4 ($t\,\delta(z) = \delta(tz)$ for $t \in [0,1]$)**.**
The final step can be completed using continuity of the metric. Now let $t \in [0,1]$ and $t_n \in \mathbb{Q}$ with $t_n \to t$ for $n \to \infty$. Then

$$\begin{aligned}
\delta(tz) &= \delta\left(\lim_{n\to\infty} t_n z\right) \\
&= \lim_{n\to\infty} \delta(t_n z) && \text{(continuity)} \\
&= \lim_{n\to\infty} t_n\,\delta(z) && (t_n \in \mathbb{Q} \text{ and part 3}) \\
&= t\,\delta(z)
\end{aligned}$$

□

5.1.1 Convexity of δ Balls

In this subsection we will proof the following lemma which is needed to conclude the proof of lemma 5.7. Everything will be assumed as in the lemma 5.7. Note, that all topological terms are with respect to the natural topology of the euclidean space.

Lemma 5.11 (Convexity of δ-Spheres)**.**
If $\mathrm{B}_\delta[\alpha] \subset S$ *is bounded, then it is convex.*

First, we will state some definitions and results on basic convex geometry that we need, and prove the theorem at the end of the section. The definitions and results on convex geometry are taken out of (Hug & Weil, 2020).

First of all, we need the notions of convex combination, the convex hull and extreme points. A convex combination is similar to a linear combination, but has the additional restraint that the coefficients must be positive ad add up to 1. The convex hull of a set is, like any other hulls, the smallest convex set that contains the set.

Definition 5.12 (Convex Combination and Hull)**.**

1. Let $k \in \mathbb{N}$, let $x_1, \dots, x_k \in \mathbb{R}^n$, and let $a_1, \dots, a_k \in [0,1]$ be such that $a_1 + \dots + a_k = 1$. Then $a_1 x_1 + \dots + a_k x_k$ is called a *convex combination* of the points $x_1, \dots x_k$.

2. For $A \subset \mathbb{R}^n$ the *convex Hull* $\mathrm{conv}(A)$ is the smallest convex set containing A, i.e.

$$\mathrm{conv}(A) := \cap \{C \subset \mathbb{R}^n : A \subset C, C \text{ convex}\}$$

The next theorem sums up the relationship between convex combination and convex hulls.

Theorem 5.13.
For $A \subset \mathbb{R}^n$ *the convex Hull is the set of all convex combination of points in* A*:*

$$\mathrm{conv}(A) = \left\{ \sum_{i=1}^{k} a_i x_i : k \in \mathbb{N}, x_1, \dots, x_k \in A, a_1, \dots, a_k \in [0,1] : \sum_{i=1}^{k} a_i = 1 \right\}$$

Moreover, the convex hull preserves compactness, as the following theorem shows.

Theorem 5.14 (convex Hull of compact set is compact)**.**
If $A \subset \mathbb{R}^n$ *is compact, then* $\mathrm{conv}(A)$ *is compact.*

Next, we will introduce extreme points, which are, in a sense, characteristic points of a set that cannot be represented by a (strict) convex combination.

Definition 5.15 (Extreme Point)**.**
Let $A \subset \mathbb{R}^n$ be closed and convex. A point $x \in A$ is called an *extreme point of* A if x cannot be represented as a nontrivial (or strict) convex combination of points of A, that is, if $x = a \cdot y + (1-a) \cdot z$ with $y, z \in A$ and $a \in (0,1)$, implies that $x = y = z$. The set of all extreme points of A is denoted by $\mathrm{extreme}(A)$.

For compact and convex sets, the extreme points are enough to "build" the convex hull of the set by doing convex combinations. In particular, this means that for a compact and convex set, every point can be expressed as a convex combination of extreme points.

Theorem 5.16 (Minkowski Theorem)**.**
Let $A \subset \mathbb{R}^n$ *be compact and convex. Then* $A = \operatorname{conv}(\operatorname{extreme} A)$*.*

Before moving to the convexity of balls, we will first prove two lemma, which we will need while proving convexity of spheres. The first one is the openness/closedness of δ-balls.

Lemma 5.17 (δ-Balls are open/closed)**.**
For every radius $r > 0$*, the* δ*-Balls are open* $(\mathrm{B}_\delta(r))$ *or closed* $(\mathrm{B}_\delta[r])$*.*

Proof. $\delta : S' \times S' \to \mathbb{R}_{\geq 0}$ is continuous with $S' \subseteq \mathbb{R}^n$ and therefore $y \mapsto \delta(y) = \delta(0,\ y)$ is continuous.

Consider the closed interval $U' = [0, \alpha] \subset \mathbb{R}_{\geq 0}$. Then

$$\begin{aligned}\delta^{-1}(U') &= \{y \in S' : \delta(y) \in U'\} \\ &= \{y \in S' : \delta(0,\ y) \leq \alpha\} \\ &= \mathrm{B}_\delta[\alpha]\end{aligned}$$

Due to topological continuity of δ (which is equivalent to preimages of closed sets being closed) $\mathrm{B}_\delta[\alpha]$ is closed. By translation invariance this holds for closed balls with any center in S. Todo: ref topo stetig? Moreover, we can obtain the same result for open balls by repeating this proof and replacing the closed interval with an open one and $\leq$ with $<$. $\square$

The second lemma shows, that extreme points of balls get compressed by a factor of $\alpha \in \mathbb{N}$, if the radius of the ball gets shrinked by the same factor.

Lemma 5.18 (compressing extreme points)**.**
Let $\alpha \in \mathbb{R}_{>0}$ *and* $r \in \mathbb{N}_{>0}$*. Further let* $\mathrm{E}(\alpha) := \operatorname{extreme}(\mathrm{B}_\delta[\alpha])$ *and* $\mathrm{C}[\alpha] := \operatorname{conv}(\mathrm{B}_\delta[\alpha])$*. Then,* $y \in \mathrm{E}(\alpha)$ *implies* $\frac{1}{r}y \in \mathrm{E}\left(\frac{\alpha}{r}\right)$*.*

Proof. We will prove this statement by contraposition.

Assume, that $\frac{1}{r}y$ is not an extreme point, i.e. $\frac{1}{r}y \in \mathrm{C}\left[\frac{\alpha}{r}\right] \setminus \mathrm{E}\left(\frac{\alpha}{r}\right)$. Then we can write $\frac{1}{r}y$ as a strict convex combination of elements $w^i \in \mathrm{C}\left[\frac{\alpha}{r}\right]$:

$$\frac{1}{r}y = \sum_{i=1}^{p} \lambda_i w^i \quad \text{with} \quad \sum_{i=1}^{p} \lambda_i = 1, \quad \lambda_i > 0,\ w^i \in \mathrm{C}\left[\frac{\alpha}{r}\right] \text{ and } p \in \mathbb{N}$$

But because $\mathrm{C}\left[\frac{\alpha}{r}\right] = \operatorname{conv}(\mathrm{B}_\delta\left[\frac{\alpha}{r}\right])$ holds, we can write any element from $\mathrm{C}\left[\frac{\alpha}{r}\right]$ as a convex combination of elements from $\mathrm{B}_\delta\left[\frac{\alpha}{r}\right]$ and therefore we can assume $w^{(i)} \in \mathrm{B}_\delta\left[\frac{\alpha}{r}\right]$.

By translation invariance and the triangle inequality we get for all i:

$$\begin{aligned}
\delta\left(rw^{(i)}\right) &\leq \delta\left(0,\ w^{(i)}\right) + \delta\left(w^{(i)},\ rw^{(i)}\right) && \text{triangle inequality}\\
&= \delta\left(w^{(i)}\right) + \delta\left(0,\ (r-1)\cdot w^{(i)}\right) && \text{translation invariance}\\
&\leq \delta\left(w^{(i)}\right) + \delta\left(0,\ w^{(i)}\right) + \delta\left(w^{(i)},\ (r-1)\cdot w^{(i)}\right) && \text{triangle inequality}\\
&= 2\cdot\delta\left(w^{(i)}\right) + \delta\left(0,\ (r-2)\cdot w^{(i)}\right) && \text{translation invariance}\\
&\dots\\
&\leq r\cdot\delta\left(w^{(i)}\right)\\
&= r\cdot\frac{\alpha}{r} = \alpha\\
\Longrightarrow\ rw^{(i)} &\in \mathrm{B}_\delta\left[\alpha\right]
\end{aligned}$$

This yields, that $y = \sum_{i=1}^{p} \lambda_i r w^{(i)}$ is a strict convex combination (because $\lambda_i r \neq 0$) and therefore y is not in $\mathrm{E}\left(\alpha\right)$. □

Now we can prove the convexity of δ-balls.
Todo: ref heine-borel?

Proof of lemma 5.11. Let $\mathrm{E}\left(\alpha\right) := \mathrm{extreme}(\mathrm{B}_\delta\left[\alpha\right])$ and $\mathrm{C}\left[\alpha\right] := \mathrm{conv}(\mathrm{B}_\delta\left[\alpha\right])$. By lemma 5.17 $\mathrm{B}_\delta\left[\alpha\right]$ is closed and by assumption it is also bounded. Therefore, from the Heine-Borel Theorem, it follows that $\mathrm{B}_\delta\left[\alpha\right]$ is compact.

By definition and theorem 5.14 it follows, that $\mathrm{C}\left[\alpha\right]$ is compact (and therefore closed, which is the reason why we denote it with square brackets) and convex. Application of the Minkowski Theorem (theorem 5.16) yields:

$$\mathrm{C}\left[\alpha\right] = \mathrm{conv}(\mathrm{extreme}(\mathrm{C}\left[\alpha\right]))$$

Further, it holds that $\mathrm{extreme}(\mathrm{C}\left[\alpha\right]) = \mathrm{extreme}(\mathrm{B}_\delta\left[\alpha\right])$ (By doing convex combinations, no new extreme points can be introduced because these are the points, which cannot be represented by a convex combination). Combining this and the statement above yields:

$$\mathrm{B}_\delta\left[\alpha\right] \subset \mathrm{C}\left[\alpha\right] = \mathrm{conv}(\mathrm{extreme}(\mathrm{B}_\delta\left[\alpha\right])).$$

Now, if we prove that $\mathrm{conv}(\mathrm{extreme}(\mathrm{B}_\delta\left[\alpha\right])) \subset \mathrm{B}_\delta\left[\alpha\right]$, we get that $\mathrm{B}_\delta\left[\alpha\right] = \mathrm{C}\left[\alpha\right]$ which proves the convexity of $\mathrm{B}_\delta\left[\alpha\right]$. We simplify this statement before proving it.

$$\mathrm{conv}(\underbrace{\mathrm{extreme}(\mathrm{B}_\delta\left[\alpha\right])}_{\mathrm{E}(\alpha)}) \subset B$$

$$\Updownarrow$$

$$\sum_{i=1}^{p} \lambda_i y^{(i)} \in \mathrm{B}_\delta\left[\alpha\right], \quad p \in \mathbb{N}_{>0}, \begin{cases} y^{(i)} \in \mathrm{E}\left(\alpha\right) \\ \lambda_i \in \mathbb{R}_{\geq 0}, \sum_{i=1}^{p} \lambda_i = 1 \end{cases} \qquad \text{(Definition)}$$

$$\Uparrow$$

$$\sum_{i=1}^{p} \lambda_i y^{(i)} \in \mathrm{B}_\delta\left[\alpha\right], \quad p \in \mathbb{N}_{>0}, \begin{cases} y^{(i)} \in \mathrm{E}\left(\alpha\right) \\ \lambda_i \in \mathbb{Q}_{\geq 0}, \sum_{i=1}^{p} \lambda_i = 1 \end{cases} \qquad (B \text{ closed, } \mathbb{Q} \text{ dense in } \mathbb{R}\ ^4)$$

$$\Uparrow$$

$$\sum_{i=1}^{r} \frac{1}{r} y^{(i)} \in \mathrm{B}_\delta\left[\alpha\right], \quad r \in \mathbb{N}_{>0},\ y^{(i)} \in \mathrm{E}\left(\alpha\right) \qquad (r \text{ common denominator})$$

Thus, we will continue proving that for $r \in \mathbb{R}_{>0}$ and $y^{(i)} \in \mathrm{E}\left(\alpha\right)$ it holds that $\sum_{i=1}^{r} \frac{1}{r} y^{(i)} \in \mathrm{B}_\delta\left[\alpha\right]$.

For $y^{(i)} \in \mathrm{E}\left(\alpha\right)$ it holds by lemma 5.18 that $\frac{1}{r} y^{(i)} \in \mathrm{E}\left(\frac{\alpha}{r}\right) \subset \mathrm{B}_\delta\left[\frac{\alpha}{r}\right]$. This means, that for the metric distance of the convex combination the following holds:

$$\begin{aligned} \delta\left(0,\ \sum_{i=1}^{r} \frac{1}{r} y^{(i)}\right) &\leq \delta\left(0,\ \frac{1}{r} y^{(1)}\right) + \delta\left(\frac{1}{r},\ \sum_{i=1}^{r} \frac{1}{r} y^{(i)}\right) && \text{(triangle inequality)} \\ &= \delta\left(0,\ \frac{1}{r} y^{(1)}\right) + \delta\left(0,\ \sum_{i=2}^{r} \frac{1}{r} y^{(i)}\right) && \text{(translation invariance)} \\ &\leq \frac{\alpha}{r} + \delta\left(0,\ \sum_{i=2}^{r} \frac{1}{r} y^{(i)}\right) && \text{(translation invariance)} \\ &\dots && \text{(repeat)} \\ &\leq r \frac{\alpha}{r} = \alpha \end{aligned}$$

This yields that $\sum_{i=1}^{r} \frac{1}{r} y^{(i)} \in \mathrm{B}_\delta\left[\alpha\right]$ holds which proves the statement. $\square$

[4] Density implies, that we can express every real number as a limit of rational numbers. B is closed, and therefore all limits of rational coefficients are included, if the statement holds for rational numbers.

Theorem 5.19 (Eindeutigkeitssatz zu l^p Metriken).
Suppose $\langle A, \succsim \rangle$ is a complete, segmentally additive, factorial proximity structure, that is also and additive difference structure.
Then there exists a unique $r \geq 1$ and real-valued functions φ_i, defined on A_i, $i = 1, \ldots, n$, that are unique up to a positive linear transformation, such that, for all $a, b, c, d \in A$,

$$(a, b) \succsim (c, d) \iff \delta(a, b) \geq \delta(c, d),$$

where

$$\delta(a, b) = \left[\sum_{i=1}^{n} |\varphi_i(a) - \varphi_i(b)|^r \right]^{1/r}.$$

Proof. Der Beweis passiert in mehreren Schritten. Zuerst sagen wir etwas über die Definitions- und Zielbereiche der Funktionen F_i und finden damit Grenzen für den Definitionsbereich von F. Danach argumentieren wir, dass wir F bis auf die positiven reellen Zahlen $[0, \infty)$ erweitern können und dass diese Erweiterung eine Funktionsgleichung impliziert, die nur von $f(x) = kx^p$ erfüllt wird. Im letzten Schritt zeigen wir, dass F auf ganz $\mathbb{R}^+$ gleich der Erweiterung sein muss und damit δ eine l^p Metrik sein muss.

Todo: Übersicht vom Beweis aufschreiben

$$F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] \tag{7}$$

Todo: necessary sufficient stuff benutzen und zeigen, dass wir das Lemma nutzen können!

Part 1.
By theorem 5.1, the additive difference model

$$\delta(a, b) = G\left(\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right)$$

yields a metric (that is additive on the coordinate axes) if $F_i(\alpha) = F(t_i\alpha)$ and $F = G^{-1}$ for $i = 1, ..., n$. Thus, F^{-1} is defined for all numbers $\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)$, and each F_i coincides with F on its domain (except for the factor t_i, which scales the argument and adjusts the domain).
Since the domains of F_i are intervals (because δ is a metric with additive segments) Todo: begründung checken , we can choose $\omega > 0$ such that for each $i = 1, \ldots, n$ and any α with $0 \leq \alpha \leq \omega/t_i$, α lies within the domain of F_i.
In particular, if we set $\omega_i = \omega/t_i$, then $F^{-1}\left[\sum_{i=1}^{n} F_i(\omega_i)\right] = F^{-1}\left[nF(\omega)\right] =: \Omega$ is defined. Thus, $F = G^{-1}$ is defined at least on the interval $[0, \Omega]$. We can now extend F as follows.

Part 2.
For $\alpha < \omega$, we have by eq. (7),

$$\begin{aligned} F^{-1}[nF(\alpha)] &= F^{-1}[nF(\alpha\omega/\omega)] \\ &= (\alpha/\omega)F^{-1}[nF(\omega)] \\ &= \alpha\Omega/\omega. \end{aligned}$$

Thus F satisfies the following, by plugging in the correct values for α ($\tilde{\alpha} = \alpha\frac{\omega}{\Omega}$ and $\tilde{\alpha} = F^{-1}(\frac{\alpha}{n})$)

$$F(a) = nF(a\omega/\Omega), \quad 0 \le a \le \Omega; \tag{8}$$

$$F^{-1}(a) = (\Omega/\omega)F^{-1}(a/n), \quad 0 \le a \le nF(\omega). \tag{9}$$

These two equations can now be used to extend the domain of F and F^{-1}.
We use eq. (8) by defining F through the right-hand side (for values where the right-hand side is defined, i.e., for $0 \le a \le \Omega^2/\omega$). Since $\Omega^2/\omega > \Omega$, this indeed extends F beyond the interval $[0, \Omega]$.
Similarly, F^{-1} satisfies equation eq. (9) for the extended interval $0 \le a \le nF(\Omega) = n^2F(\omega)$.
Moreover, eq. (7) is now valid for $0 \le t\alpha_i \le \Omega$:

$$\begin{aligned} F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} nF(t\alpha_i\omega/\Omega)\right] && \text{(by eq. (8))} \\ &= F^{-1}\left[n\sum_{i=1}^{n} F(t\alpha_i\omega/\Omega)\right] \\ &= \frac{\Omega}{\omega}F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i\omega/\Omega)\right] && \text{(by eq. (9))} \\ &= \frac{\Omega}{\omega}tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i\omega/\Omega)\right] \\ & \qquad \text{(by homogeneity applied for } \alpha_i\omega/\Omega \le \omega) \\ &= tF^{-1}\left[n\sum_{i=1}^{n} F(\alpha_i\omega/\Omega)\right] && \text{(by eq. (9))} \\ &= tF^{-1}\left[\sum_{i=1}^{n} nF(\alpha_i\omega/\Omega)\right] \\ &= tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] && \text{(by eq. (8))} \end{aligned}$$

Repeatedly applying the arguments, extends the intervals in which F is defined and in which homogeneity (eq. (7)) is valid by a factor of Ω/ω each time. Thus, we can extend F to $[0, \infty)$ while preserving homogeneity for $0 \le t \le 1$.

Part 3.

The function obtained through this extension must therefore coincide with the original F on at least the interval $[0, \Omega)$. Furthermore, applying eq. (9) on the mean-value function $F^{-1}\left[\left(\frac{1}{n}\sum_{i=1}^{n} F(\alpha_i)\right)\right]$ yields:

$$F^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(\alpha_i)\right]. \tag{10}$$

According to (Hardy et al., 1952, thm. 84), the only monotonic functions satisfying this equation for $0 \le \alpha < \infty$ are functions of the form $k\alpha^r$ (see also Aczél, 1966, S. 153).

Therefore, $F(\alpha) = k\alpha^r$ must hold on at least the interval $[0, \Omega]$. Now, for any $|\varphi_i(a_i) - \varphi_i(b_i)|$, we can find $t > 0$ small enough such that $t|\varphi_i(a_i) - \varphi_i(b_i)| < \omega, i = 1, \ldots, n$. Applying monotonicity yields (we can assume $t \le 1$):

$$\begin{aligned}
\delta(a,b) &= F^{-1}\left(\sum_{i=1}^{n} F\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t}F^{-1}\left(\sum_{i=1}^{n} F\left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t}\left(\sum_{i=1}^{n} \left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)^r\right)^{1/r} && (\text{ since } t|\varphi_i(\alpha_i) - \varphi_i(\beta_i)| \le \omega) \\
&= \left(\sum_{i=1}^{n} |\varphi_i(a_i) - \varphi_i(b_i)|^r\right)^{1/r}
\end{aligned}$$

Thus, δ is an l^p metric and, by Theorem Todo: ref add-diff-prox-struct-repr , $F = k \cdot \alpha^r$ and hence $F_i = k(t_i\alpha)^r$ or $F_i(\alpha) = k\alpha^r$ if we incorporate the t_i into the φ_i. Finally, the restriction $1 \le r < \infty$ follows from the triangle inequality or, more precisely, from Minkowski's inequality.

□

6 Discussion

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